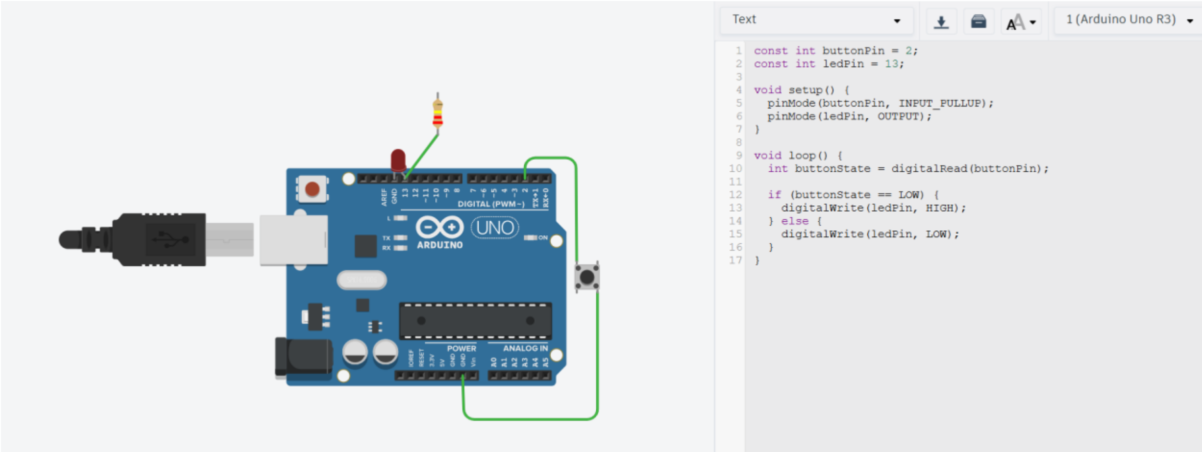
UTHM

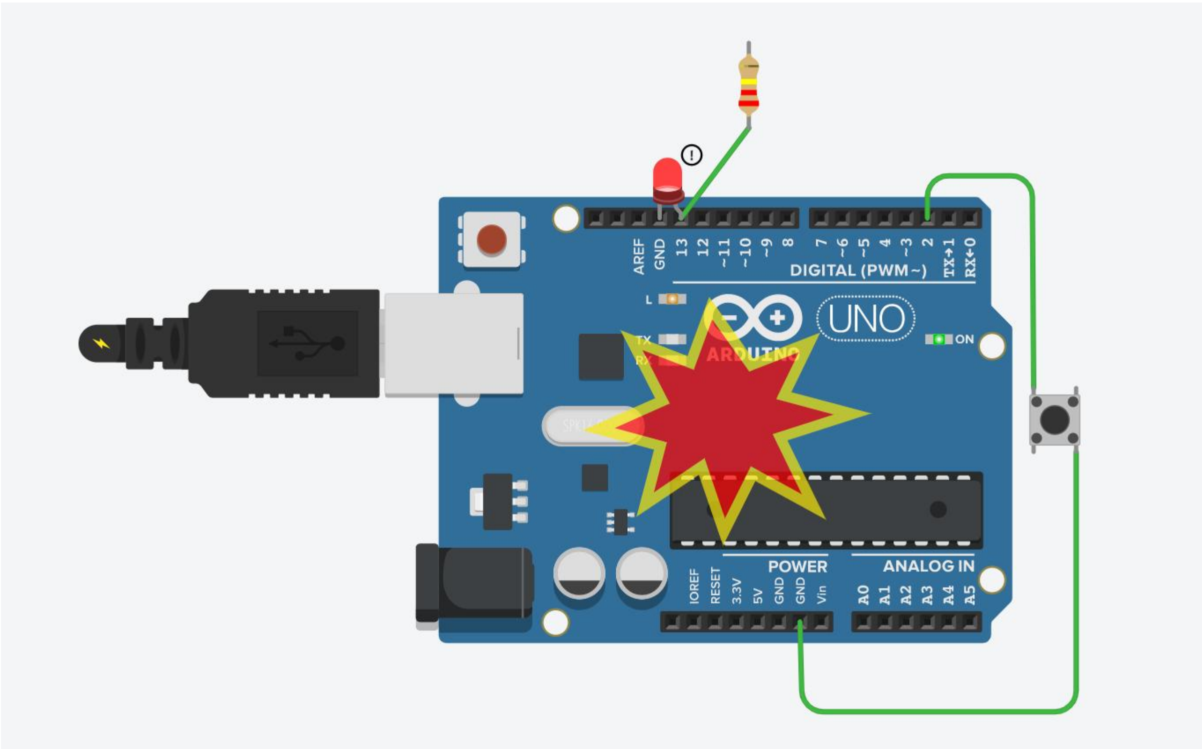
Universiti Tun Hussein Onn Malaysia

BIW33803 – IoT

Lab 1 – Exploring Arduino Virtual Simulator with Tinkercad

Student Name	Muhammad Daniel Iman Bin Mohamad Redha
Matric Number	CI230046
Section	S5





Review Question

a. What is Tinkercad?

Tinkercad is a web-based design and simulation platform developed by Autodesk that allows users to create 3D models, design electronic circuit and simulate Arduino projects without physical hardware. It is widely used for learning electronics and programming in a virtual environment.

b. What is the difference between the Circuits and Code menus in Tinkercad?

- **Circuits menu**: Used to build and connect electronic components such as Arduino boards, LEDs, resistors and buttons.
- **Code menu**: Used to write and edit programs that control how the circuit behaves.

c. How do you create a circuit in Tinkercad?

To create a circuit:

1. Open Tinkercad and go to the Circuits section.
2. Click “Create new Circuit”.
3. Drag and drop components.
4. Connect components using wires.
5. Ensure proper connections.

d. How do you write code for an Arduino in Tinkercad?

To write Arduino code:

1. Open the circuit project.
2. Click the “Code” button.
3. Switch to Text mode (Arduino IDE).
4. Write the program using Arduino syntax.
5. Run the simulation to test the code.

e. What is the purpose of the virtual simulator/LAB environment in Tinkercad?

The virtual simulator allows users to test circuits and code without physical components. It helps in:

- Debugging wiring and logic errors
- Understanding how circuits behave
- Reducing cost and risk of damaging real hardware

Reflection

During this lab activity, I learned how to design and simulate a simple Arduino circuit using Tinkercad, including connecting components such as an LED and a push button and writing code to control them. I also gained a better understanding of digital input and output, particularly how a button can be used to control an LED.

One of the main challenges I encountered was incorrect wiring, especially with the button and LED connections. At first, the LED did not light up because the circuit was incomplete or incorrectly connected. Another challenge was understanding how the INPUT_PULLUP mode works, as it requires a different wiring approach compared to using external resistors.

To overcome these challenges, I carefully reviewed the circuit connections, tested different configurations and referred to correct wiring methods. Using the simulation feature also helped me identify mistakes quickly and fix them without needing physical hardware.